Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Spatial AI–Guided Cell Isolation Platform for Cancer Organoid–Based Biomarker Discovery			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Pharmaceutical Bio-based Technologies and Systems	Bio-processes/Bio-equipment	Bio-engineering Technologies
Concept of Proposed Technology	The proposed technology is an **integrated translational discovery platform** that links **spatial molecular profiling, AI-guided pathology, and spatially guided precision cell isolation** to enable robust target and biomarker discovery from **patient-derived tissues and patient-derived organoid models**. Conventional discovery workflows typically rely on dissociated or bulk tissue analysis, resulting in loss of spatial context and dilution of rare but disease-driving cell populations. In contrast, the proposed platform treats **spatial context as a core analytical dimension**, enabling identification of *where disease biology occurs* and direct recovery of the exact cells responsible for disease progression, resistance, or degeneration. Key functional components include: • **Spatial molecular profiling** of tissues and organoids to generate region-resolved molecular signatures			

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	• **AI-assisted pathology** to identify and prioritize disease-relevant regions and cell populations • **Spatially-resolved Laser Activated Cell Sorting (SLACS)-enabled spatially guided cell isolation**, allowing targeted recovery of rare and spatially restricted cells • **Deep molecular profiling and network analysis** of isolated cells to identify actionable targets and biomarkers • **Functional validation in patient-derived organoids**, enabling causal assessment of candidate targets The platform produces **translation-ready outputs**, including disease-specific spatial atlases, target–biomarker packages, organoid-based assays, and standardized workflows suitable for joint R&D and downstream commercialization.
Rationale for Joint R&D Project	**Unmet Need** Many high-burden diseases are driven by **spatially localized, interaction-dependent, and rare cell states** (e.g., invasive tumor fronts, immune–tissue interfaces, fibrotic microdomains). While spatial profiling technologies can identify these regions, a major bottleneck remains: the inability to **reliably isolate the exact disease-relevant cells** from patient tissues and organoids for downstream molecular and functional validation. This gap delays translation from spatial observation to **drug target nomination and clinically testable biomarker hypotheses**. **Rationale for International Collaboration** The joint R&D structure is essential because: • A **leading U.S. academic clinical research environment** provides access to clinically annotated patient tissues, disease-diverse cohorts, pathology expertise, and patient-derived organoid systems necessary for global translational relevance. • The **Korean SME partner owns SLACS**, a proprietary spatially guided cell isolation technology that cannot be replicated or substituted using standard laboratory methods and is the critical bridge from spatial data to isolatable cell populations. • The value of the project emerges from an **iterative co-development loop** (spatial analysis → SLACS isolation → profiling → organoid validation → refinement), which cannot be achieved through outsourcing or unilateral execution. Together, the collaboration transforms spatial biology from a descriptive mapping exercise into a **target discovery and validation engine** aligned with global drug discovery standards.
Key Words	1. Spatially Resolved Cell Sorting 2. SLACS-Enabled Precision Cell Isolation 3. Cancer Patient-Derived Tissue and Organoid Analysis 4. Translational Target Discovery Platform 5. Context-Aware Single-Cell Profiling

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)

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R&D Objectives	**Overall Objectives** To develop and validate an **end-to-end translational discovery platform** integrating AI pathology, spatial molecular profiling, SLACS-enabled cell isolation, and patient-derived organoid validation, and to advance the technology from **TRL 3 to TRL 6**. **Specific Objectives** • Develop AI-guided pathology tools for reproducible identification of disease-relevant regions • Advance SLACS workflows for precise isolation of spatially prioritized cell populations • Integrate isolated-cell profiling with patient-derived organoid functional validation • Establish standardized, reproducible workflows suitable for multi-site deployment • Generate clinically grounded target and biomarker discovery outputs

R&D Target Specifications	**AI Pathology and Spatial Prioritization**
	
	• Region identification accuracy: **≥ 85%** concordance with expert pathologists
	• False-positive rate: **≤ 15%**
	• Processing time per whole-slide image: **≤ 5 minutes**
	• Inter-run reproducibility: **≥ 90%** region overlap
	
	SLACS-Enabled Cell Isolation
	
	• Target cell recovery rate: **≥ 70%**
	• Cell purity: **≥ 90%**
	• Minimum recoverable cell number: **100–300 cells per region**
	• Downstream assay compatibility:
	  ○ RNA profiling success ≥ **95%**
	  ○ Low-input multi-omics success ≥ **90%**
	
	Tissue and Organoid Applicability
	
	• Validated patient tissue samples: **≥ 50**
	• Organoid establishment success rate: **≥ 70%**
	• Tissue–organoid signature concordance: **≥ 70%**
	
	Discovery Outputs
	
	• Candidate targets identified: **≥ 10–15 per disease area**
	• Functionally validated targets: **≥ 3–5**
	• Spatial biomarkers: **≥ 5 per disease area**
	• Cross-patient reproducibility: **≥ 30–40%** patient prevalence

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will serve as the **clinical and translational research lead**, including: • Provision of clinically annotated patient tissues and pathology expertise • Generation and maintenance of patient-derived organoid models • Validation of AI pathology outputs through expert review • Execution of functional validation studies in cancer organoids • Contribution to translational interpretation and dissemination
Expected R&R of Korean SME Partner	The Korean SME partner will serve as the **core technology owner and platform development lead**, including: • Development, optimization, and deployment of SLACS technology • Engineering of spatially guided cell isolation workflows • Definition of isolation QC metrics and SOPs • Integration of SLACS outputs with profiling and analysis pipelines • Support for reproducibility testing and technology transfer

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Impact** • Establishment of a **validated spatial-to-target translational discovery platform** • Demonstration of SLACS in a globally credible clinical research environment • Creation of standardized, reproducible workflows for spatial target discovery **Commercial and Industrial Impact** • Positioning of SLACS as a globally applicable core spatial biology technology • Enablement of commercialization via instruments, software, reagents, and workflow kits • Expansion opportunities for Korean SMEs in automation, AI, and bioinformatics **Societal and Strategic Impact**

	• Acceleration of human-relevant target and biomarker discovery • Improved translation from patient biology to therapeutic development • Strengthening of U.S.–Korea biotechnology collaboration models

6. References

1. Amos C. Lee, Sumin Lee, Sunghoon Kwon; Abstract 6781: Spatial omics using spatially-resolved laser-activated cell sorting for cancer biomarker discovery. *Cancer Res* 1 April 2023; 83 (7_Supplement): 6781. https://doi.org/10.1158/1538-7445.AM2023-6781
2. Ho, W.J., Jaffee, E.M. & Zheng, L. The tumour microenvironment in pancreatic cancer — clinical challenges and opportunities. *Nat Rev Clin Oncol* **17**, 527–540 (2020). https://doi.org/10.1038/s41571-020-0363-5
3. Su CY, Burchett A, Dunworth M, Choi JS, Ewald AJ, Ahn EH, Kim DH. Engineering a 3D collective cancer invasion model with control over collagen fiber alignment. Biomaterials. 2021 Aug;275:120922. doi: 10.1016/j.biomaterials.2021.120922. Epub 2021 Jun 4. PMID: 34126408; PMCID: PMC8450056.